

MIDDLE PENINSULA, VA, USA

WMO: 720285

Lat: 37.521N

Lon: 76.765W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.9	(c) -5.9	(d) -19.8	(e) 0.6	(f) -3.2	(g) -16.3	(h) 0.9	(i) -1.6	(j) 10.0	(k) 10.9	(l) 8.4	(m) 9.8	(n) 1.6	(o) 320	(p) 0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.0	25.6	33.4	24.8	32.3	24.3	27.3	32.0	26.5	30.8	25.8	29.9	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.4	25.2	20.4	28.6	24.7	19.7	28.0	85.7	31.4	82.4	30.6	79.6	30.1	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.8	(b) 6.7	(c) 5.6	(e) -13.0	(f) 37.0	(g) 4.5	(h) 1.4	(i) -16.2	(j) 38.0	(k) -18.8	(l) 38.8	(m) -21.3	(n) 39.7	(o) -24.6	(p) 40.7		
			DB													
			WB													
				-13.7	28.0	4.0	1.9	-16.6	29.4	-18.9	30.5	-21.2	31.5	-24.1	32.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.1	5.5	9.2	15.2	20.2	24.2	26.6	25.6	22.4	16.3	9.9	6.2
	DBStd	9.16	5.78	5.48	5.51	4.92	4.31	3.33	2.56	2.63	3.38	4.74	4.97	5.38
	HDD10.0	654	199	149	85	10	0	0	0	0	0	5	61	145
	HDD18.3	1980	442	361	286	120	32	2	0	0	7	97	256	378
	CDD10.0	2662	16	22	61	166	316	427	516	483	371	201	58	28
	CDD18.3	947	0	1	4	26	90	178	257	224	127	33	4	2
	CDH23.3	9077	1	4	42	269	753	1749	2722	2189	1042	244	47	16
CDH26.7	3544	0	0	5	71	248	692	1204	891	346	54	26	7	
Wind	WSAvg	2.2	2.3	2.5	2.7	2.8	2.3	2.0	1.9	1.7	1.8	2.0	2.0	2.0
Precipitation	PrecAvg	1141	86	69	106	85	97	94	118	113	117	88	78	90
	PrecMax	1649	161	157	221	172	198	150	215	345	404	180	226	227
	PrecMin	879	30	16	6	34	23	40	34	32	6	0	5	28
	PrecStd	187	33	38	46	32	44	33	57	70	92	49	50	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	23.4	27.3	31.3	32.9	36.2	37.5	36.3	34.1	31.0	28.9	24.2
		MCWB	16.5	16.9	17.0	19.6	22.6	25.4	25.5	26.8	23.3	22.4	23.9	18.8
	2%	DB	18.9	20.9	24.0	28.7	31.3	33.8	35.1	33.9	32.2	27.9	22.9	20.5
		MCWB	15.3	14.6	15.7	18.6	22.1	24.8	25.9	25.2	23.8	21.7	17.8	17.8
	5%	DB	16.5	17.8	21.7	26.6	29.4	32.2	33.5	32.5	30.2	26.2	21.1	17.8
		MCWB	13.0	12.8	14.5	17.9	21.4	24.0	25.4	24.8	22.8	20.5	16.9	15.6
	10%	DB	13.7	15.0	18.9	24.1	27.7	30.8	32.3	31.2	28.6	24.0	18.3	15.0
		MCWB	10.4	10.6	13.6	16.8	20.7	23.4	24.9	24.1	22.1	19.7	15.0	12.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.2	19.5	21.9	24.9	27.2	28.4	28.3	26.3	25.0	23.4	20.7
		MCDB	20.4	21.9	23.2	26.2	30.4	31.8	34.0	33.7	29.7	27.5	26.5	22.1
	2%	WB	16.0	16.4	17.8	20.7	23.7	26.2	27.3	27.2	25.3	23.7	19.3	18.0
		MCDB	18.5	19.0	21.7	25.7	28.3	30.6	32.5	31.6	28.5	26.4	21.5	19.7
	5%	WB	13.7	13.9	16.2	19.4	22.8	25.4	26.6	26.4	24.6	22.5	17.6	15.9
		MCDB	16.4	17.4	20.0	24.6	27.1	30.2	31.3	30.2	27.7	24.7	20.0	17.7
	10%	WB	10.8	11.0	14.4	18.2	21.9	24.6	25.9	25.6	23.8	20.7	15.7	13.0
		MCDB	12.9	14.0	18.3	22.6	25.9	29.0	30.2	29.2	27.0	23.4	17.8	14.5
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.2	11.9	12.6	11.2	10.9	10.2	10.3	10.8	11.3	11.9	10.6
		MCDBR	13.7	14.5	14.9	14.7	12.1	11.9	11.1	11.2	12.3	12.6	13.1	12.7
		MCWBR	10.6	9.6	8.5	6.5	4.9	4.7	4.2	4.2	5.0	6.6	8.4	10.2
	5% WB	MCDBR	13.0	13.8	12.6	12.4	10.4	10.4	10.0	9.7	9.4	9.6	11.0	12.3
		MCWBR	11.2	10.7	9.0	6.5	5.1	4.5	4.4	4.5	4.7	5.7	8.7	10.7
Clear-Sky Solar Irradiance	taub	0.305	0.317	0.348	0.392	0.431	0.485	0.515	0.511	0.410	0.354	0.322	0.309	
	taud	2.532	2.490	2.439	2.327	2.249	2.144	2.080	2.076	2.344	2.500	2.544	2.555	
	Ebn at Noon	888	918	916	888	855	806	778	770	839	864	859	859	
	Edh at Noon	77	91	105	125	137	153	162	159	115	90	76	71	
All-Sky Solar Radiation	RadAvg	2.30	3.06	4.07	5.14	5.84	6.34	6.05	5.46	4.43	3.47	2.60	2.03	
	RadStd	0.18	0.34	0.33	0.38	0.55	0.44	0.30	0.27	0.42	0.40	0.22	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (75 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	+100

Nomenclature: See separate page